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DAY-2(SEPT 3)Class Work

Day-2: File Ownership and Access Control

Topics: chown, chgrp, ownership + permissions

Part A — Ownership Investigation (10 Minutes)

Create a directory named ownership_lab.

- mkdir ownership_lab
- cd ownership_lab

Inside it, create:

incident_report.txt

network.log

credentials.txt

```
anonymous@anonymous-supreme in ~/ownership_lab
) touch incident_report.txt network.log credentials.txt

anonymous@anonymous-supreme in ~/ownership_lab
) ls -l
-rw-r--r-- 0 anonymous 3 Sep 11:11 credentials.txt
-rw-r--r-- 0 anonymous 3 Sep 11:11 incident_report.txt
-rw-r--r-- 0 anonymous 3 Sep 11:11 network.log
```

Display the detailed information for these files.

For each file, identify:

Owner

Group

Owner permissions

Group permissions

Others permissions

```
anonymous@anonymous-supreme in ~/ownership_lab
# ) ls -l incident_report.txt network.log credentials.txt
-rw-r--r-- 0 anonymous 3 Sep 11:11 credentials.txt
-rw-r--r-- 0 anonymous 3 Sep 11:11 incident_report.txt
-rw-r--r-- 0 anonymous 3 Sep 11:11 network.log
anonymous@anonymous-supreme in ~/ownership_lab
```

Determine the UID and GID of the current user.

```
anonymous@anonymous-supreme in ~/ownership_lab
# ) id
uid=1000(anonymous) gid=1000(anonymous) groups=1000(anonymous),3(sys),96(scan
ner),108(vboxusers),209(cups),944(docker),953(sambashare),970(realtime),985(v
ideo),991(lp),994(input),998(wheel)
anonymous@anonymous-supreme in ~/ownership_lab
```

Compare the UID/GID information with the ownership displayed for the newly created files.

```
anonymous@anonymous-supreme in ~/ownership_lab
[*] # stat incident_report.txt
File: incident_report.txt
Size: 0          Blocks: 0          IO Block: 4096   regular empty fil
e
Device: 0,48     Inode: 5460826    Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/anonymous)   Gid: ( 1000/anonymous)
Access: 2026-09-03 11:11:06.887068577 +0530
Modify: 2026-09-03 11:11:06.887068577 +0530
Change: 2026-09-03 11:11:06.887068577 +0530
Birth: 2026-09-03 11:11:06.887068577 +0530
```

Explain the difference between file ownership and file permissions.

- **File ownership** who owns the file and which group is associated with it.
- **File permission** determine what the owner, group member, and everyone else are allowed to do with the file

Part B — chown: Changing Ownership (15 Minutes)

Change the owner of incident_report.txt to alice.

```
anonymous@anonymous-supreme in ~/ownership_lab took 27s396ms
# ) sudo chown alice incident_report.txt
[sudo] password for anonymous:

anonymous@anonymous-supreme in ~/ownership_lab as 🧑 took 4s472ms
# ) ls -l incident_report.txt
-rw-r--r-- 0 alice 3 Sep 11:11 incident_report.txt
```

Change the owner of network.log to bob.

```
anonymous@anonymous-supreme in ~/ownership_lab as 🧑
# ) sudo chown bob network.log

anonymous@anonymous-supreme in ~/ownership_lab as 🧑
# ) ls -l network.log
-rw-r--r-- 0 bob 3 Sep 11:11 network.log
```

Change the owner of credentials.txt to charlie.

Verify each change.

```
anonymous@anonymous-supreme in ~/ownership_lab as 🧑
# ) sudo chown charlie credentials.txt

anonymous@anonymous-supreme in ~/ownership_lab as 🧑
# ) ls -l credentials.txt
-rw-r--r-- 0 charlie 3 Sep 11:11 credentials.txt
```

Try changing the owner of a file to another user without using sudo. Record the result.

```
anonymous@anonymous-supreme in ~/ownership_lab
) touch test_ownership.txt

anonymous@anonymous-supreme in ~/ownership_lab
) chown bob test_ownership.txt
chown: changing ownership of 'test_ownership.txt': Operation not permitted
```

Explain why an ordinary user is generally not permitted to transfer file ownership to another user.

- A normal user can't just give their file to someone else — the system won't allow it, mainly for security reasons. Only root can change file ownership freely.

Using a single command, configure:

incident_report.txt

Owner : alice

Group : security

Verify the resulting ownership.

```
anonymous@anonymous-supreme in ~/ownership_lab
[*] ) sudo chown alice:security incident_report.txt
[sudo] password for anonymous:

anonymous@anonymous-supreme in ~/ownership_lab as  took 7s426ms
) ls -l incident_report.txt
-rw-r--r-- 0 alice  3 Sep 11:11 incident_report.txt

anonymous@anonymous-supreme in ~/ownership_lab as  )
```

Part C — chgrp: Changing Group Ownership (15 Minutes)

Change the group ownership of network.log to developers.

```
anonymous@anonymous-supreme in ~/ownership_lab as 🐼  
# ) sudo chgrp developers network.log  
  
anonymous@anonymous-supreme in ~/ownership_lab as 🐼  
# ) ls -l network.log  
-rw-r--r-- 0 bob 3 Sep 11:11 network.log
```

Change the group ownership of credentials.txt to security.

Verify the changes.

```
anonymous@anonymous-supreme in ~/ownership_lab as 🐼  
# ) sudo chgrp security credentials.txt  
  
anonymous@anonymous-supreme in ~/ownership_lab as 🐼  
# ) ls -l credentials.txt  
-rw-r--r-- 0 charlie 3 Sep 11:11 credentials.txt  
  
anonymous@anonymous-supreme in ~/ownership_lab as 🐼
```

Display the group memberships of alice, bob, and charlie.

```
anonymous@anonymous-supreme in ~/ownership_lab as 🐼  
# ) id alice id bob id charlie  
uid=1001(alice) gid=1001(alice) groups=1001(alice),1004(security)  
id: 'id': no such user  
uid=1002(bob) gid=1002(bob) groups=1002(bob),1004(security)  
id: 'id': no such user  
uid=1003(charlie) gid=1003(charlie) groups=1003(charlie),1004(security)  
  
anonymous@anonymous-supreme in ~/ownership_lab as 🐼
```

Log in as alice and create a file named alice_report.txt.

```
anonymous@anonymous-supreme in ~/ownership_lab as 🐼  
[*] # sudo -iu alice  
-bash: n: command not found  
[alice@anonymous-supreme ~]$ whoami  
alice  
[alice@anonymous-supreme ~]$ id  
uid=1001(alice) gid=1001(alice) groups=1001(alice),1004(security)  
[alice@anonymous-supreme ~]$ _
```

Attempt to change the group of `alice_report.txt` to security.

```
anonymous@anonymous-supreme in ~/ownership_lab took 12m47s844ms
) touch alice_report.txt

anonymous@anonymous-supreme in ~/ownership_lab
) ls -l alice_report.txt
-rw-r--r-- 0 anonymous 3 Sep 17:09 alice_report.txt
```

Attempt to change its group to developers.

```
anonymous@anonymous-supreme in ~/ownership_lab
) chgrp security alice_report.txt
chgrp: changing group of 'alice_report.txt': Operation not permitted

anonymous@anonymous-supreme in ~/ownership_lab
[*] ) ls -l alice_report.txt
-rw-r--r-- 0 anonymous 3 Sep 17:09 alice_report.txt
```

Compare the results.

```
anonymous@anonymous-supreme in ~/ownership_lab
) chgrp developers alice_report.txt
chgrp: changing group of 'alice_report.txt': Operation not permitted

anonymous@anonymous-supreme in ~/ownership_lab
[*] ) _
```

Explain how a user's group membership affects their ability to change the group ownership of a file they own.

- A user who owns a file can generally change its group to a group that the user is a member of

Part D — Ownership + Permission Investigation (20 Minutes)

Configure:

`-rw-r-----` alice security incident_report.txt

```
anonymous@anonymous-supreme in ~/ownership_lab took 4s302ms
[*] ) sudo chown alice:security incident_report.txt
[sudo] password for anonymous:

anonymous@anonymous-supreme in ~/ownership_lab as  took 4s657ms
) sudo chmod 640 incident_report.txt

anonymous@anonymous-supreme in ~/ownership_lab as  took 4s657ms
) ls -l incident_report.txt
-rw-r----- 0 alice 3 Sep 11:11 incident_report.txt
```

Identify what operations are available to:

alice

charlie

bob

A user who belongs to neither security nor the file owner

Deleting a file is controlled by the permission of the directory containing the file, not the file's own read/write permissions

User	Relationship	Read	Write	Delete
Alice	Owner	yes	yes	Depends on directory
Charlie	Member of security	yes	no	Depends on directory
Bob	If not security	no	no	Depends on directory
Neither	Others	no	no	Depends on directory

charlie belongs to security. Test whether Charlie can:

Read incident_report.txt

Modify incident_report.txt

Delete incident_report.txt

Record the result of each operation.

```
anonymous@anonymous-supreme in ~/ownership_lab as 🐼  
➤ ) sudo -iu charlie  
-bash: n: command not found  
[charlie@anonymous-supreme ~]$ cd /home/anonymous/ownership_lab  
[charlie@anonymous-supreme ownership_lab]$ whoami  
pwd  
charlie  
/home/anonymous/ownership_lab  
[charlie@anonymous-supreme ownership_lab]$ cat incident_report.txt  
[charlie@anonymous-supreme ownership_lab]$ ^C  
[charlie@anonymous-supreme ownership_lab]$ cat incident_report.txt  
[charlie@anonymous-supreme ownership_lab]$ echo "Charlie modified this" >> incident_report.txt  
-bash: incident_report.txt: Permission denied  
[charlie@anonymous-supreme ownership_lab]$ rm incident_report.txt  
rm: remove write-protected regular empty file 'incident_report.txt'?  
[charlie@anonymous-supreme ownership_lab]$
```

Change the permissions to:

-rw-rw----

```
anonymous@anonymous-supreme in ~/ownership_lab as 🐼 took 3m51s368ms
➤ ) sudo chmod 660 incident_report.txt

anonymous@anonymous-supreme in ~/ownership_lab as 🐼
➤ ) ls -l incident_report.txt
-rw-rw---- 0 alice 3 Sep 11:11 incident_report.txt
```

Repeat the test as charlie.

```
➤ ) sudo -iu charlie
-bash: n: command not found
[charlie@anonymous-supreme ~]$ cat ~/ownership_lab/incident_report.txt
cat: /home/charlie/ownership_lab/incident_report.txt: No such file or directory
[charlie@anonymous-supreme ~]$ cd /home/anonymous/ownership_lab
[charlie@anonymous-supreme ownership_lab]$ cat ~/ownership_lab/incident_report.txt
cat: /home/charlie/ownership_lab/incident_report.txt: No such file or directory
[charlie@anonymous-supreme ownership_lab]$ pwd
/home/anonymous/ownership_lab
[charlie@anonymous-supreme ownership_lab]$ cat incident_report.txt
[charlie@anonymous-supreme ownership_lab]$ echo "Charlie test" >> incident_report.txt
[charlie@anonymous-supreme ownership_lab]$ cat incident_report.txt
Charlie test
[charlie@anonymous-supreme ownership_lab]$ echo "Charlie test" >> incident_report.txt
[charlie@anonymous-supreme ownership_lab]$
```

Explain why Charlie's access changed even though the owner and group of the file did not change.

- Charlie is a member of security, so the group permission changed from read-only to read + write.

Remove bob from the security group.

```
anonymous@anonymous-supreme in ~/ownership_lab took 8m16s456ms
➤ ) sudo gpasswd -d bob security
[sudo] password for anonymous:
Removing user bob from group security

anonymous@anonymous-supreme in ~/ownership_lab as 🐼 took 9s995ms
➤ ) id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob)
```

Without changing the file's ownership or permissions, test Bob's access again.

```

anonymous@anonymous-supreme in ~/ownership_lab as 🐼
➤ ) sudo -iu bob
-bash: n: command not found
[Bob@anonymous-supreme ~]$ cat ~/ownership_lab/incident_report.txt
cat: /home/bob/ownership_lab/incident_report.txt: No such file or directory
[Bob@anonymous-supreme ~]$ cd /home/anonymous/ownership_lab
[Bob@anonymous-supreme ownership_lab]$ cat ~/ownership_lab/incident_report.tx
t
cat: /home/bob/ownership_lab/incident_report.txt: No such file or directory
[Bob@anonymous-supreme ownership_lab]$ cat incident_report.txt
cat: incident_report.txt: Permission denied
[Bob@anonymous-supreme ownership_lab]$ echo "Bob test" >> incident_report.txt
-bash: incident_report.txt: Permission denied
[Bob@anonymous-supreme ownership_lab]$ _

```

Explain why Bob's effective permissions changed.

Bob's effective permission changed because his membership in the file's group was removed. The file's owner, group and permission bits remained unchanged. Since Bob was no longer a member of security, he lost the group's "rw" permission and instead received the "--" permission assigned to others.